The Measurement Problem in Darke Theory: Partial Structure, Closure, and Registrational Commitment

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Abstract

The quantum measurement problem arises from the tension between continuous unitary evolution and the appearance of definite recorded outcomes. Standard quantum mechanics permits superposed evolution, while measurement yields persistent classical results. Decoherence explains suppression of interference between branches, but does not by itself fully explain why one outcome becomes the realised registered result. This paper develops a Darke-theoretic account of the problem by distinguishing three structural levels: formal availability, partial structural admissibility, and persistent realised structure. Within this framework, quantum states are interpreted as partially structured regimes near the closure boundary, where multiple continuations may remain admissible without yet being fully committed into a single persistent branch. Closure is treated as a general structural process and not as something generated solely by measurement or by observers. Measurement is instead interpreted as an additional bridge across the closure threshold, especially on the side of durable registration. A closure-bearing registration frame couples to the partially structured state and helps carry one admissible branch into persistent recorded commitment. The observer is therefore not treated psychologically or metaphysically, but structurally, as any system capable of stable registration, memory retention, identity-preserving propagation, and downstream consistency enforcement. The paper then develops a formal core in which compatibility is grounded in event-token identity, common downstream record extension, and non-fusability of incompatible record values; bounded support is derived from finite record-bearing capacity; and exclusive measurement outcomes are shown to yield persistent registrational singularity in any finite-support closure-bearing frame satisfying identity-preserving propagation and event-token uniqueness. A quantum specialisation is given in terms of orthogonal apparatus record states, stable downstream propagation, and record-forming system–apparatus–environment interaction. On this view, measurement is neither primitive magical collapse nor the sole cause of reality, but a specialised

closure-selection event in which unresolved multiplicity ceases to remain jointly supportable as persistent recorded structure.

1 Introduction

The quantum measurement problem remains one of the deepest unresolved fractures in modern physics. Quantum theory is extraordinarily successful as a predictive formalism. It describes interference, spectra, transition probabilities, entanglement, and correlated outcomes with extraordinary precision. Yet its predictive success does not by itself settle the ontological status of measurement.

The formal difficulty is familiar. A quantum system evolves according to linear unitary dynamics, preserving superposition. But measurements do not present themselves as superpositions. They present themselves as definite, durable, classically legible outcomes. The formalism therefore appears to move from a regime of multiplicity to a regime of singularity, yet the structural meaning of that transition remains contested.

This paper develops a Darke-theoretic account of that transition. The central proposal is not that observers create reality, not that consciousness collapses the wavefunction, and not that measurement is the sole source of closure. The stronger and more disciplined claim is different. Closure is a general structural process operating throughout physical reality. Measurement is one specialised case in which a partially structured regime is carried across a threshold into durable registrational commitment. Measurement is therefore interpreted as an additional bridge across the closure threshold, not as the sole engine of closure.

The proposal turns on a distinction that standard discussions often compress too quickly. There is a difference between:

(1) formal availability,

(2) partial structural admissibility,

(3) persistent registered realisation.

A mathematical description may permit multiple continuations. Those continuations may remain structurally admissible in a partially formed regime. Yet it does not follow that all such continuations are jointly maintainable as persistent recorded structure. The core claim of the present paper is that quantum superposition belongs to this middle regime: partial structure, not yet fully closure-committed and not yet durably registered.

This distinction permits a cleaner statement of the measurement problem. The strongest question is not simply, "How does an observer make one branch real?" The deeper question is:

> Under what conditions does a formally available multiplicity cease to remain jointly admissible as persistent recorded structure?

This is the correct Darke-theoretic entry point.

The paper has seven aims:

(i) to restate the measurement problem in its strongest standard form;

(ii) to introduce the Darke-theoretic distinction between partial and persistent structure;

(iii) to define closure-bearing registration frames in non-psychological terms;

(iv) to formulate measurement as an additional bridge across the closure threshold;

(v) to explain why one outcome becomes persistently registered;

(vi) to situate the proposal relative to decoherence, collapse, and many-worlds;

(vii) to establish the structural theorem architecture by which jointly persistent registration of incompatible exclusive outcomes is excluded under explicit persistence axioms.

The central thesis of the paper may be stated in one sentence:

> Measurement does not generate closure from nothing; it acts as an additional bridge by which partially structured admissible states are carried into persistent registered structure.

2 The Measurement Problem in Standard Form

The standard quantum measurement problem arises because the theory appears to involve two very different kinds of state transition.

On the one hand, quantum evolution is linear and unitary. If a state vector $\psi(t)$ evolves under a Hamiltonian $\hat{H}$, then

$$i\hbar \frac{d}{dt}\psi(t) = \hat{H}\psi(t),$$

or equivalently by a unitary propagator. This structure preserves superposition.

On the other hand, measurement yields definite outcomes. If a state is expanded in an eigenbasis,

$$\psi = \sum_j c_j \phi_j,$$

then the formalism associates weights $|c_j|^2$ with possible outcomes, but actual measurement does not present itself as a lived superposition of all outcomes. It presents a single recorded result.

The wound can be stated sharply:

(a) the quantum formalism carries multiple continuations forward;

(b) measurement produces one definite record;

(c) the transition from the first regime to the second is not fully settled by bare unitary dynamics alone.

This gives rise to the familiar unresolved questions:

- What physically counts as measurement?
- Why does one result appear rather than an unresolved multiplicity?
- What is the ontological status of the non-realised terms?
- Why does classical definiteness emerge where it does?
- Is collapse real, effective, emergent, epistemic, or unnecessary?

Primitive collapse models answer by adding a new fundamental process. Decoherence-based accounts explain suppression of interference and emergence of effectively diagonal reduced descriptions, but do not by themselves fully settle why one outcome becomes the persistent recorded result. Many-worlds preserves formal multiplicity but leaves unresolved the distinction between mathematical branching and persistent record-bearing realised structure. Observer-centred accounts become unstable when "observation" is left psychologically vague.

The structural diagnosis advanced here is that these debates are often distorted by failure to distinguish:

(i) what is mathematically available,

(ii) what is partially but not yet fully structurally committed,

(iii) what is persistently realised and durably recorded.

The present paper treats the measurement problem as a threshold problem between these layers.

3 Foundational Setting in Darke Theory

The present construction stands downstream of the Darke-theoretic sequence

$$\text{recursive differentiation} \Longrightarrow \text{closure} \Longrightarrow \text{admissibility} \Longrightarrow \text{persistent structure.}$$

Only the minimum load-bearing points are required here.

3.1 Recursive differentiation

Darke Theory does not begin from primitive objecthood. It begins from distinction, refinement, and progressive structural differentiation. Stable structure is not given at the start.

3.2 Closure

Differentiation alone does not produce a stable world. Structure must become self-integrating and self-supporting. Closure is the process by which proliferating structure becomes persistent structure.

3.3 Admissibility

Not everything that can be formally described can be persistently carried. Admissibility is the filter separating bare formal possibility from genuinely supportable structure.

3.4 Persistent structure

What is real in the strong structural sense is not merely what can be written down, but what can be stably borne, preserved, and propagated.

3.5 Importance for measurement

This changes the shape of the quantum measurement problem. The right question is not "How does an observer magically force one outcome to occur?" The stronger question is:

> How does a formally available multiplicity fail to remain jointly admissible as persistent recorded structure?

That is the correct Darke-theoretic entry point.

4 Core Definitions

Definition 4.1 (Formal availability)**.** *A formally available state is a state or continuation admitted by the governing mathematical description, regardless of whether it is yet stably realised as persistent structure.*

Definition 4.2 (Partially structured state)**.** *A partially structured state is a state whose admissible continuations are not yet uniquely closure-committed into a single persistently realised and registered branch.*

Definition 4.3 (Closure)**.** *Closure is the structural process by which a regime becomes sufficiently self-integrating, self-supporting, and stabilised to persist as coherent realised structure.*

Definition 4.4 (Closure threshold)**.** *The closure threshold is the boundary at which a state ceases to be merely conditionally or partially admissible and becomes supportable as stable persistent structure.*

Definition 4.5 (Registration)**.** *Registration is the durable embedding of a state into a wider closure-bearing network of memory, record, and downstream consistency.*

Definition 4.6 (Registrational commitment)**.** *Registrational commitment is the transition by which one admissible branch becomes persistently embedded in a record-bearing structure such that it is preserved, propagated, and made externally legible.*

Definition 4.7 (Registration threshold)**.** *The registration threshold is the boundary at which a state becomes durably fixed within a wider closure-bearing system of records, memory, and consistency constraints.*

Definition 4.8 (Closure-bearing registration frame)**.** *A closure-bearing registration frame is any physical system capable of preserving outcome, retaining memory, preserving record identity under propagation, and enforcing downstream consistency across interacting degrees of freedom.*

Definition 4.9 (Measurement event)**.** *A measurement event is an interaction in which a partially structured state is coupled to a closure-bearing registration frame such that one admissible branch is carried into persistent registrational commitment.*

Definition 4.10 (Closure-selection)**.** *Closure-selection is the narrowing of unresolved admissible multiplicity into one persistently realised and registered branch under closure, burden, and registration constraints.*

Definition 4.11 (Residual branch structure)**.** *Residual branch structure denotes formally representable but non-carried alternatives that remain part of the unresolved pre-registration description yet are not jointly maintained as persistent recorded reality.*

5 Three Structural Levels

The distinction between three structural levels is the conceptual centre of the paper.

5.1 Formal availability

A branch may be present in the mathematical description. This is the weakest level. It tells us only that the branch belongs to the admissible formal representation.

5.2 Partial structural admissibility

A stronger regime exists between mathematics and classical realised structure. In this middle regime, multiple continuations are not mere fiction, but neither are they yet fully committed as persistent outcome records. This is the regime of partial structure.

Without this middle layer, one is forced into a crude binary:

- either all branches are equally fully real;
- or one branch becomes real by a primitive unexplained collapse.

The Darke-theoretic distinction shows that a third option exists: multiple continuations can be genuinely part of the state without yet being jointly persistent as realised recorded structure.

5.3 Persistent registered realisation

At the strongest level, a branch is closure-bearing, durably recorded, and embedded in downstream consistency. This is the level of apparatus outcomes, macroscopic records, stable memory traces, and classical persistence.

Principle 5.1 (Structural stratification)**.** *A formally available branch, a partially admissible branch, and a persistently realised registered branch are not identical kinds of being. A theory that conflates them obscures the real hinge of the measurement problem.*

6 Closure Is Broader Than Measurement

Closure is not caused solely by measurement. Closure is a general structural process operating throughout physical reality. Systems may approach or cross closure thresholds

through intrinsic dynamics, environmental coupling, admissibility narrowing, or other stabilising interactions.

Stars form without observers. Crystals stabilise without observers. Black holes collapse without observers. Molecular environments decohere local states without observers in any psychological sense.

Measurement is therefore not the sole source of closure.

Principle 6.1 (Closure is broader than measurement)**.** *Closure is a general structural process and is not generated solely by observers, measurements, or acts of awareness.*

This principle is non-negotiable. Without it, the framework would collapse into a weak idealism that the present theory explicitly rejects.

7 Closure-Bearing Registration Frames

The language of "observer" is too loose unless reinterpreted. The correct structural object is the closure-bearing registration frame.

What matters is not subjective awareness, but whether a system can:

(i) preserve an outcome,

(ii) retain a record,

(iii) preserve record identity,

(iv) propagate downstream consistency,

(v) embed the result into a wider closure-bearing network.

Any system that does this can function as a closure-bearing registration frame.

Examples include:

- a detector plate,
- a measuring apparatus,
- a decohering environmental record structure,
- a biological memory system,
- a human brain.

The last of these is therefore not metaphysically unique. It is one member of a broader structural class.

Principle 7.1 (Observer de-psychologised)**.** *The observer is not privileged as consciousness. What matters is the existence of a closure-bearing registration frame capable of preserving outcome, preserving record identity, and enforcing downstream consistency.*

8 Measurement as an Additional Bridge Across the Threshold

We can now state the core thesis.

Principle 8.1 (Additional threshold-bridging)**.** *Measurement is an additional bridge by which a partially structured state is carried across the closure threshold into persistent registered structure.*

Every word here matters.

8.1 Why "additional" matters

The word *additional* is essential. It preserves the fact that:

- closure is broader than measurement,
- systems can close without observers,
- measurement still has a real ontological role,
- but only as one bridge among others.

8.2 Measurement acts most strongly at the registrational side

Not every closure event is a measurement event. Measurement is special because it produces durable records. It therefore acts most strongly at the registrational side of the threshold: the point where one branch becomes externally fixed, preserved, and legible in a wider system.

8.3 Measurement is not passive readout

On this view, pre-measurement structure is not simply a hidden classical fact waiting to be passively revealed. The state is only partially structured. Measurement participates in the transition into registrational commitment.

8.4 Measurement is not primitive magic

What appears as collapse is reinterpreted as threshold-crossing from partial admissibility into persistent recorded structure under closure-bearing interaction.

9 Registration as Ontological Carriage

A hostile ambiguity must be removed at once: the paper's notion of registration is not epistemic bookkeeping. It is ontological carriage.

Principle 9.1 (Ontological registration)**.** *A record in the present paper is not merely a description-relative or observer-relative bookkeeping state. It is a physically carried persistent structure embedded in memory, identity-preserving propagation, and downstream consistency constraints.*

This point matters because the main theorem concerns not merely what can be said about a system, but what can be jointly carried as realised persistent record structure.

10 Notation Summary

For clarity, we collect here the principal symbols used in the formal core.

Symbol	**Meaning**
X	Class of admissible structural states.
$B(x)$	Branch family of formally available continuations of state $x \in X$.
F	Closure-bearing registration frame.
$\mathcal{N}(F)$	Class of downstream record networks extending frame F.
$\mathcal{E}$	Event-token space.
$\mathcal{V}$	Record-value space.
$\mathrm{Adm}(b \mid x)$	Local admissibility indicator for branch b from state x.
$\mathrm{Cl}(b \mid x)$	Closure depth of branch b relative to state x.
$\mathrm{Reg}(b \mid x, F)$	Registrational embedding strength of branch b in frame F.
$\Pi_F(x)$	Registrational partiality functional of state x relative to frame F.
C_{crit}	Closure threshold.

R_{crit}	Registration threshold.
$B_F^{\text{reg}}(x)$	Set of branches individually eligible for persistent registration in frame F.
$\text{Rec}_F(b)$	Record state induced in frame F by branch b.
$\text{Evt}_F(b)$	Event token assigned to branch b in frame F.
$\text{Val}_F(b)$	Record value assigned to branch b in frame F.
$\text{Comp}_F(b_i, b_j)$	Compatibility of branches b_i, b_j in frame F.
$\text{Inc}_F(b_i, b_j)$	Incompatibility of branches b_i, b_j in frame F.
$\beta(b \mid x, F)$	Base burden of sustaining branch b in frame F.
λ	Irreducible contradiction penalty for pairwise incompatible registrations.
$\text{Bur}(B \mid x, F)$	Burden of jointly carrying branch family B in frame F.
$S_{\text{Id}}, S_{\text{Mem}}, S_{\text{Prop}}, S_{\text{Cons}}$	Component support capacities.
$S(x, F)$	Total finite support bound of frame F at state x.
T_F	Additional bridging operator induced by coupling state x to frame F.

11 Formal Core: States, Branches, Records, and Thresholds

The preceding sections state the conceptual architecture. We now construct a stronger formal core in which compatibility, burden, support, and outcome singularity are tied to explicit structural conditions.

11.1 State space and branch family

Let X denote the class of admissible structural states. For each state $x \in X$, let

$$B(x)$$

denote the set of formally available branch continuations of x.

No Hilbert-space-specific assumption is made at this stage. The present construction is intended as a structural layer that can later be specialised to ordinary quantum state evolution.

Definition 11.1 (Branch family)**.** *For each state $x \in X$, the branch family $B(x)$ is the set of continuations formally available from x according to the governing description.*

11.2 Admissibility, closure, and registration functionals

For each branch $b \in B(x)$ and each closure-bearing registration frame $F \in \mathcal{F}$, define

$$\mathrm{Adm}(b \mid x) \in \{0, 1\}, \qquad \mathrm{Cl}(b \mid x) \in [0, \infty), \qquad \mathrm{Reg}(b \mid x, F) \in [0, \infty).$$

We distinguish two thresholds:

$$C_{\mathrm{crit}} > 0, \qquad R_{\mathrm{crit}} > 0.$$

Definition 11.2 (Closed branch)**.** *A branch $b \in B(x)$ is structurally closed relative to x if*

$$\mathrm{Adm}(b \mid x) = 1 \quad \textit{and} \quad \mathrm{Cl}(b \mid x) \geq C_{\mathrm{crit}}.$$

Definition 11.3 (Registrationally committed branch)**.** *A branch $b \in B(x)$ is registrationally committed in frame F if*

$$\mathrm{Adm}(b \mid x) = 1, \qquad \mathrm{Cl}(b \mid x) \geq C_{\mathrm{crit}}, \qquad \mathrm{Reg}(b \mid x, F) \geq R_{\mathrm{crit}}.$$

Remark 11.4. *This distinction is load-bearing. Closure is broader than measurement. Measurement does not generate closure in general; it contributes specifically to registrational crossing where durable records are formed.*

11.3 Partial structure and the partiality functional

We now sharpen the middle regime mathematically.

Definition 11.5 (Registrational partiality)**.** *For a state x relative to frame F, define*

$$\Pi_F(x) := \sum_{b \in B(x)} \mathrm{Adm}(b \mid x) \max\left(0,\, 1 - \frac{\mathrm{Reg}(b \mid x, F)}{R_{\mathrm{crit}}}\right).$$

Definition 11.6 (Partially structured state, sharpened form)**.** *A state $x \in X$ is partially structured relative to a relevant class of frames if:*

(i) there exist distinct admissible branches $b_1, b_2 \in B(x)$ with

$$\mathrm{Adm}(b_1 \mid x) = \mathrm{Adm}(b_2 \mid x) = 1,$$

and

(ii) for at least one relevant record-bearing frame F, one has

$$\Pi_F(x) > 0.$$

This strengthens the earlier threshold-only notion. Partial structure is now marked by nonzero unresolved registrational partiality, not only by descriptive prose.

11.4 Registrable branch set

For a fixed frame F, define

$$B_F^{\mathrm{reg}}(x) := \{b \in B(x) : \mathrm{Adm}(b \mid x) = 1,\ \mathrm{Cl}(b \mid x) \geq C_{\mathrm{crit}},\ \mathrm{Reg}(b \mid x, F) \geq R_{\mathrm{crit}}\}\,.$$

This is the set of branches individually eligible for persistent registration in F.

12 Compatibility from Event Tokens, Record Values, and Common Extension

Compatibility must not remain a black-box consistency slogan. It must be grounded in event tokens, record values, and identity-preserving downstream extension.

12.1 Record networks, event tokens, and record values

Let $\mathcal{N}(F)$ denote the class of downstream record networks extending frame F. An element $N \in \mathcal{N}(F)$ is a persistence-bearing extension of F containing memory, record propagation, and consistency constraints downstream of the measurement interaction.

For each branch $b \in B_F^{\mathrm{reg}}(x)$, define:

$$\mathrm{Rec}_F(b), \qquad \mathrm{Evt}_F(b) \in \mathcal{E}, \qquad \mathrm{Val}_F(b) \in \mathcal{V}.$$

These are interpreted as:

- $\mathrm{Rec}_F(b)$: the record induced in frame F by branch b;
- $\mathrm{Evt}_F(b)$: the event token to which the record belongs;
- $\mathrm{Val}_F(b)$: the settled record value assigned to that event token.

For downstream extension into N, the induced record must preserve event-token identity and record-value attribution under propagation.

12.2 Non-fusability of incompatible settled values

Definition 12.1 (Non-fusability)**.** *Two record values $v_i, v_j \in \mathcal{V}$ are non-fusible for event token $e \in \mathcal{E}$ if they cannot be jointly realised as one settled value assignment for e within any identity-preserving realised record network.*

This is the primitive anti-contradiction notion the paper actually needs.

12.3 Common extension and compatibility

Definition 12.2 (Common downstream extension)**.** *Let $b_i, b_j \in B_F^{\mathrm{reg}}(x)$. A common downstream extension for (b_i, b_j) exists over F if there exists some $N \in \mathcal{N}(F)$ such that:*

(i) both records extend into N,

(ii) event-token identity is preserved under extension,

(iii) if $\mathrm{Evt}_F(b_i) = \mathrm{Evt}_F(b_j)$, then the propagated values do not violate non-fusability of settled event-value assignment.

Definition 12.3 (Compatibility and incompatibility)**.** *For $b_i, b_j \in B_F^{\mathrm{reg}}(x)$, define*

$$\mathrm{Comp}_F(b_i, b_j) = 1 \iff \exists\, N \in \mathcal{N}(F) \text{ admitting a common downstream extension for } (b_i, b_j).$$

Define incompatibility by

$$\mathrm{Inc}_F(b_i, b_j) := 1 - \mathrm{Comp}_F(b_i, b_j).$$

Remark 12.4. *This is sharper than the earlier formulation. Incompatibility is now tied not merely to a vague failure of consistency, but to identity-preserving failure of common extension for event-token/value assignments.*

12.4 Exclusive outcome families

Definition 12.5 (Exclusive outcome family)**.** *A finite set*

$$B = \{b_1, \ldots, b_n\} \subseteq B_F^{\mathrm{reg}}(x)$$

is an exclusive outcome family in frame F if:

(i) all branches in B are assigned to the same event token:

$$\mathrm{Evt}_F(b_i) = \mathrm{Evt}_F(b_j) \qquad \text{for all } i, j,$$

and

(ii) their settled values are pairwise non-fusible, equivalently

$$\mathrm{Inc}_F(b_i, b_j) = 1 \qquad \textit{for all } i \neq j.$$

This is the paper's formalisation of mutually exclusive measurement outcomes.

13 Burden, Support, and Finite Record Capacity

The key structural move now is to derive bounded support from finite record-bearing capacity at a more explicit level than before.

13.1 Burden decomposition

Let $B \subseteq B_F^{\mathrm{reg}}(x)$ be finite. Define

$$\mathrm{Bur}(B \mid x, F) = \mathrm{Bur}_{\mathrm{base}}(B \mid x, F) + \mathrm{Bur}_{\mathrm{conf}}(B \mid x, F),$$

where

$$\mathrm{Bur}_{\mathrm{base}}(B \mid x, F) = \sum_{b \in B} \beta(b \mid x, F),$$

and

$$\mathrm{Bur}_{\mathrm{conf}}(B \mid x, F) = \lambda \sum_{\substack{b_i, b_j \in B \\ i<j}} \mathrm{Inc}_F(b_i, b_j).$$

Interpretation:

- $\mathrm{Bur}_{\mathrm{base}}$ is the burden of sustaining branches at all;
- $\mathrm{Bur}_{\mathrm{conf}}$ is the irreducible burden introduced by pairwise incompatible event-value assignments within one realised record ontology.

13.2 Support decomposition

A closure-bearing registration frame is finite. We therefore decompose support into component capacities:

$$S(x, F) = S_{\mathrm{Id}}(x, F) + S_{\mathrm{Mem}}(x, F) + S_{\mathrm{Prop}}(x, F) + S_{\mathrm{Cons}}(x, F),$$

where:

- S_{Id} is record-identity support,
- S_{Mem} is memory-bearing support,
- S_{Prop} is downstream propagation support,
- S_{Cons} is consistency-maintenance support.

Definition 13.1 (Finite-support frame)**.** *A closure-bearing registration frame F is finite-support at state x if each of the component supports above is finite, and hence*

$$S(x, F) < \infty.$$

Proposition 13.2 (Finite-support law from finite record capacity)**.** *Let F be a closure-bearing registration frame with finite record discriminability, finite memory-bearing depth, finite identity-preserving propagation support, and finite consistency maintenance capacity. Then F is finite-support at every fixed state x.*

Proof. Each component support is finite by hypothesis. Since the total support $S(x, F)$ is their finite sum, it is finite. Joint persistent registration requires drawing on these component supports. Therefore the total supportable burden of any jointly persistent registered family is bounded above by $S(x, F)$. □

Definition 13.3 (Joint persistence criterion)**.** *A finite branch family $B \subseteq B_F^{\mathrm{reg}}(x)$ is jointly persistent in frame F if*

$$\mathrm{Bur}(B \mid x, F) \leq S(x, F).$$

Supportability has now formally replaced mere describability.

14 The Additional Bridging Operator

The current paper's central verbal claim is that measurement is an additional bridge across the threshold. We now formalise that claim more strongly.

Let

$$T_F : X \to X$$

denote the transition induced by coupling state x to closure-bearing registration frame F.

Axiom 14.1 (Monotonicity of registrational bridging)**.** *For every branch $b \in B(x)$,*

$$\mathrm{Cl}\Big(b \mid T_F(x)\Big) \geq \mathrm{Cl}(b \mid x),$$

and

$$\mathrm{Reg}\Big(b \mid T_F(x), F\Big) \geq \mathrm{Reg}(b \mid x, F),$$

with strict increase in the registration component whenever the interaction with F genuinely forms record-bearing structure.

Remark 14.2. *This is the formal meaning of "additional bridge". The operator T_F need not generate closure ex nihilo. It need only act monotonically on closure and nontrivially on registration where record-bearing interaction is present.*

15 Theorem Chain for Persistent Registrational Singularity

We now derive the formal spine of the paper.

15.1 Incompatibility lower bound

Lemma 15.1 (Incompatibility lower bound)**.** *Let $B \subseteq B_F^{\mathrm{reg}}(x)$ be finite. Then*

$$\mathrm{Bur}(B \mid x, F) \geq \lambda \sum_{\substack{b_i, b_j \in B \\ i<j}} \mathrm{Inc}_F(b_i, b_j).$$

Proof. By definition,

$$\mathrm{Bur}(B \mid x, F) = \sum_{b \in B} \beta(b \mid x, F) + \lambda \sum_{i<j} \mathrm{Inc}_F(b_i, b_j).$$

Since each $\beta(b \mid x, F) \geq 0$, dropping the first term yields the result. □

15.2 Quadratic growth under exclusivity

Lemma 15.2 (Quadratic burden growth for exclusive families)**.** *Suppose $B \subseteq B_F^{\mathrm{reg}}(x)$ is an exclusive outcome family. Then*

$$\mathrm{Bur}(B \mid x, F) \geq \lambda \binom{|B|}{2}.$$

Proof. If B is exclusive, then $\mathrm{Inc}_F(b_i, b_j) = 1$ for all distinct pairs, so

$$\sum_{i<j} \mathrm{Inc}_F(b_i, b_j) = \binom{|B|}{2}.$$

Apply the previous lemma. □

15.3 Finite incompatibility bound

Theorem 15.3 (Finite incompatibility bound)**.** *Let $x \in X$ and let F be finite-support at x. Then there exists an integer $N_F(x) \geq 0$ such that no exclusive outcome family*

$$B \subseteq B_F^{\mathrm{reg}}(x)$$

with

$$|B| > N_F(x)$$

can persist jointly in frame F.

Proof. If B were jointly persistent, one would require

$$\mathrm{Bur}(B \mid x, F) \leq S(x, F).$$

By the previous lemma,

$$\lambda \binom{|B|}{2} \leq \mathrm{Bur}(B \mid x, F) \leq S(x, F).$$

Hence

$$\binom{|B|}{2} \leq \frac{S(x, F)}{\lambda}.$$

Since the left-hand side grows quadratically in $|B|$ while the right-hand side is finite, only finitely many values of $|B|$ satisfy the inequality. Let $N_F(x)$ be the largest such value. □

Corollary 15.4 (No unbounded incompatible record multiplicity)**.** *A finite-support closure-bearing registration frame cannot jointly maintain arbitrarily large families of mutually exclusive persistent outcome records.*

15.4 Identity preservation and event-token uniqueness

We now derive the singularity step from more primitive assumptions than the old slogan of single-history consistency.

Axiom 15.5 (Identity-preserving propagation)**.** *If a branch record is persistently registered in frame F, then its event token and settled record value are preserved under downstream propagation through realised record networks extending F.*

Axiom 15.6 (Event-token uniqueness)**.** *Within one realised record network, one event token cannot bear two distinct non-fusible settled record values.*

These are more primitive than simply announcing that one history cannot contain two exclusive outcomes. They state instead that identity is preserved and that one event token cannot stably carry contradictory settled values.

Proposition 15.7 (Derived single-history consistency)**.** *Let F be a closure-bearing registration frame. If identity-preserving propagation and event-token uniqueness both hold in realised record networks extending F, then F is single-history consistent in the sense that one realised record network cannot contain two distinct exclusive outcome records for one and the same measurement event.*

Proof. If two exclusive outcome records for one event token were jointly present in one realised record network, then by exclusivity their values would be non-fusible. By identity-preserving propagation the event token would remain the same under extension. This would violate event-token uniqueness. Hence no such joint realisation is possible. □

Theorem 15.8 (Persistent registrational singularity for exclusive event records)**.** *Let x be a partially structured state and let F be a closure-bearing registration frame. Suppose:*

(i) the relevant candidate outcome records form an exclusive outcome family for one event token;

(ii) identity-preserving propagation holds in realised record networks extending F;

(iii) event-token uniqueness holds in realised record networks extending F;

(iv) at least one branch in the family lies above both the closure and registration thresholds after application of the bridging operator T_F.

Then the persistent registered outcome family for that event in frame F has cardinality exactly one.

Proof. By assumption (iv), the set of persistently registrable outcomes is nonempty after application of T_F.

Suppose for contradiction that there exist two distinct branches

$$b_i, b_j \in B_F^{\mathrm{reg}}(T_F(x))$$

that persist jointly as settled records of the same event. Since the family is exclusive,

$$\mathrm{Evt}_F(b_i) = \mathrm{Evt}_F(b_j), \qquad \mathrm{Inc}_F(b_i, b_j) = 1.$$

Thus their values are non-fusible as assignments to one event token. By identity-preserving propagation, that event token remains the same under realised downstream extension. By event-token uniqueness, one realised record network cannot carry two distinct non-fusible settled values for one event token. This contradicts joint persistence. Therefore no jointly persistent outcome family can have cardinality greater than one. Since the family is nonempty, it has cardinality exactly one. □

Corollary 15.9 (Effective outcome singularity)**.** *For a closure-bearing registration frame satisfying identity-preserving propagation and event-token uniqueness, exclusive measurement yields branch-singular persistent registration.*

Remark 15.10. *Theorem 15.8 together with Corollary 15.9 constitute the main formal result of the present paper.*

This is the strongest statement in the paper. It does not claim that unregistered alternatives are metaphysically annihilated. It proves that under the stated persistence axioms they cannot be jointly carried as persistent realised records of one event within a single realised record ontology.

16 Quantum Specialisation

The formal core above is structural. We now connect its objects explicitly to ordinary quantum measurement language in order to show that the theorem-bearing architecture does not float free of the physics it is meant to interpret.

16.1 System–environment state and effective branch family

Let the total quantum system consist of a measured subsystem S, an apparatus or local registration sector A, and an environment E. Let the underlying quantum state be represented by a vector or density operator on

$$\mathcal{H}_S \otimes \mathcal{H}_A \otimes \mathcal{H}_E.$$

In measurement contexts, one typically encounters entangled states of the form

$$|\Psi\rangle = \sum_i c_i \, |s_i\rangle \otimes |a_i\rangle \otimes |e_i\rangle,$$

or the corresponding mixed-state description after partial tracing. In the present framework, the structural state

$$x \in X$$

is interpreted as an effective system–apparatus–environment configuration carrying the relevant branch structure for persistence and registration. The associated branch family

$$B(x)$$

is interpreted as the family of measurement-continuation candidates induced by such an entangled decomposition, especially after decoherence has selected a stable or approximately stable branch basis.

Remark 16.1. *The present theory does not identify branchhood with arbitrary basis expansion. The intended branch family is a physically meaningful one, associated with stable measurement continuations rather than formal basis freedom.*

16.2 Closure and registration in quantum language

In the quantum specialisation, the closure functional

$$\mathrm{Cl}(b \mid x)$$

tracks the structural stability of a branch under environment coupling, suppression of recoherence, and resistance to internal destabilisation. The registration functional

$$\mathrm{Reg}(b \mid x, F)$$

tracks the extent to which the branch is durably embedded in the apparatus/environment record sector.

The point of keeping these functionals distinct is that a branch may become dynamically stable before it becomes fully record-bearing. This mirrors the distinction between branch formation and registrational commitment already built into the structural formalism.

16.3 Event tokens and apparatus record values

Let a fixed measurement interaction be associated with an event token $e \in \mathcal{E}$. Suppose the measurement produces apparatus record states

$$|a_i\rangle, \qquad i \in I,$$

with corresponding settled value labels $v_i \in \mathcal{V}$ for the event token e.

In ordinary measurement theory, macroscopically distinguishable apparatus outcomes are represented by states that are orthogonal, or at least effectively orthogonal, at the level relevant to stable readout:

$$\langle a_i \mid a_j \rangle \approx 0 \qquad \text{for } i \neq j.$$

We interpret such orthogonality as the quantum signature of non-fusability of settled apparatus record values.

Proposition 16.2 (Orthogonal record states induce non-fusible values)**.** *Let $b_i, b_j \in B_F^{\mathrm{reg}}(x)$ correspond to apparatus record states $|a_i\rangle, |a_j\rangle$ satisfying*

$$\langle a_i \mid a_j \rangle = 0.$$

Assume that persistent registration in frame F preserves record identity and event-token assignment. Then the settled values assigned to b_i and b_j for that event token are non-fusible.

Proof. If $\langle a_i \mid a_j \rangle = 0$, then the two apparatus states represent distinct record states. If record identity and event-token assignment are preserved, then these states cannot be jointly treated as one settled value assignment for one event token without erasing record identity. Hence the values are non-fusible. □

Corollary 16.3 (Orthogonal records induce exclusivity)**.** *Under the same hypotheses, the corresponding branches form an exclusive pair in the structural sense:*

$$\mathrm{Inc}_F(b_i, b_j) = 1.$$

Proof. Non-fusible settled values for one event token violate the compatibility condition for common downstream extension. Hence $\mathrm{Comp}_F(b_i, b_j) = 0$ and therefore $\mathrm{Inc}_F(b_i, b_j) = 1$. □

This is the first direct quantum-to-structural bridge. Orthogonal apparatus records induce exclusive record ontology.

16.4 Stable downstream propagation and derived event-token uniqueness

We now connect the persistence axioms to ordinary record propagation.

A measurement outcome is not merely a local apparatus state at one instant. It is a record that propagates into downstream sectors: memory, readout, environment correlations, further causal consequences, and later descriptions. In standard physical practice, these downstream sectors preserve event attribution and record identity sufficiently well to sustain one settled outcome history for one event.

Proposition 16.4 (Stable record propagation induces the persistence axioms)**.** *Let F be a closure-bearing registration frame for a fixed measurement event. Suppose:*

(i) persistent registration in F propagates into downstream record sectors without erasing event-token identity;

(ii) persistent registration in F propagates without erasing record-value identity;

(iii) one realised record history cannot contain two distinct orthogonal apparatus records as one settled result of the same event token.

Then identity-preserving propagation and event-token uniqueness both hold for realised record networks extending F.

Proof. By hypothesis, downstream propagation preserves event-token identity and record-value identity. If two distinct orthogonal apparatus records for the same event token were jointly carried as one settled result, event-token identity would be preserved while the settled value assignment would become contradictory. This is excluded by hypothesis. Hence the persistence axioms hold. □

This is the second direct bridge. Ordinary stable measurement record practice yields the primitive persistence axioms used in the main theorem.

16.5 The bridging operator in quantum language

The structural bridging operator

$$T_F : X \to X$$

is not introduced as a substitute for unitary dynamics, nor as a second primitive collapse law. In the quantum specialisation it is interpreted as the structural transition associated with record-forming interaction between system, apparatus, and environment.

In ordinary quantum terms, this corresponds to the passage from pre-registration coupling to stable outcome-bearing correlation. Schematically:

$$|s_i\rangle \otimes |a_0\rangle \otimes |e_0\rangle \quad \longmapsto \quad |s_i\rangle \otimes |a_i\rangle \otimes |e_i\rangle,$$

followed by amplification, environmental embedding, and stabilisation of record-bearing distinctions.

The operator T_F therefore names not a violation of unitary theory, but the structural lift by which closure and registration increase under record-forming coupling.

Remark 16.5. *The role of T_F is not to replace Schrödinger evolution. It is to formalise the threshold-crossing effect of measurement interaction at the structural level relevant to persistent registration.*

16.6 Persistent registrational singularity in quantum language

We may now restate the strongest theorem of the paper in explicitly quantum-facing form.

Corollary 16.6 (Persistent registrational singularity for orthogonal measurement records)**.** *Let a measurement interaction generate a family of decoherence-stabilised branch candidates whose apparatus record states are pairwise orthogonal and whose downstream record*

propagation preserves event-token identity and settled-value identity. If at least one such branch is carried above both the closure and registration thresholds by the interaction with frame F, then the persistent registered outcome family for that event has cardinality exactly one.

Proof. By the orthogonal-record proposition and its corollary, the candidate branches form an exclusive outcome family. By the stable-record propagation proposition, the persistence axioms required by Theorem 15.5 hold. Apply Theorem 15.5. □

This establishes the Hilbert-space bridge required for the present structural result. It shows that the formal core of the paper can be instantiated in ordinary quantum measurement language through orthogonal record states, identity-preserving propagation, and record-forming system–apparatus–environment interaction.

16.7 Interpretive consequence

The consequence is sharp. Quantum superposition is not here interpreted as already-classical coexistence of multiple settled records. It is interpreted as partial structure: a regime in which multiple continuations remain admissible prior to full registrational commitment. Measurement then acts as an additional bridge across the threshold into persistent record.

That is the exact point at which the structural formalism rejoins the physics of measurement.

17 Measurement Sequence in Darke-Theoretic Terms

The full measurement process can now be written as a sequence of stages.

17.1 Stage I: Formal availability

The system possesses multiple mathematically available continuations.

17.2 Stage II: Partial structural admissibility

These continuations are not mere fiction. They remain partially structured and conditionally admissible.

17.3 Stage III: Coupling to a closure-bearing registration frame

The system interacts with a detector, apparatus, environment cascade, or equivalent record-capable frame. Formally, the state is acted on by T_F.

17.4 Stage IV: Registrational burden emerges

Once durable registration is at stake, jointly carrying incompatible outcomes becomes structurally expensive or inadmissible.

17.5 Stage V: Closure-selection

One branch is carried across the threshold into registrational commitment.

17.6 Stage VI: Classical persistence

The realised outcome is embedded in records, memories, environmental traces, and downstream consistency. This is classical definiteness in structural terms.

18 Relation to Decoherence

The present account does not reject decoherence. It integrates it while refusing to reduce the whole story to it.

18.1 What decoherence explains

Decoherence explains why interference becomes inaccessible, why branch isolation emerges, and why preferred bases often appear under environmental coupling.

18.2 What decoherence does not by itself settle

Decoherence does not by itself fully answer why one outcome becomes the persistent recorded result rather than merely one branch among several effectively isolated branches.

18.3 Darke-theoretic placement

In the present framework, decoherence is treated as a major part of the closure process. It suppresses interference and prepares branch isolation. Measurement through a closure-bearing registration frame provides the additional bridge into registrational commitment.

Principle 18.1 (Decoherence-plus-registration)**.** *Decoherence isolates branch structure; registrational commitment carries one branch into persistent realised record-bearing structure.*

19 Relation to Collapse Models

The present account differs from primitive collapse models.

It rejects collapse as an unexplained second law inserted into the formalism. But it does retain the appearance of collapse at the level of realised structure. Once closure-selection occurs, a single registered outcome persists.

The reinterpretation is:

> Collapse is not primitive magical deletion of alternatives. It is the structural effect of closure-selection when unresolved multiplicity can no longer remain jointly supportable as persistent recorded structure.

20 Relation to Many-Worlds

The present account neither simply denies nor simply adopts many-worlds.

It agrees that multiple continuations may be present in the formal description and may remain meaningful prior to full registrational commitment. What it resists is the slide from mathematical branching to equal persistent realised structure.

The formal theorem of this paper is therefore not a theorem of bare branch multiplicity. It is a theorem of persistent realised record ontology. Formal multiplicity is not denied. What is excluded is the joint persistent registration of incompatible exclusive outcomes within one realised record-bearing structure.

21 Classicality as Deep Closure

The framework yields a natural account of classicality.

Classicality is not a separate primitive ontological regime. It is a regime of deep closure and thick registration.

A deeply closed regime exhibits:

(i) durable memory,

(ii) robust record formation,

(iii) stable downstream consistency,

(iv) strong environmental embedding,

(v) resistance to unresolved multiplicity as jointly carried structure.

Macroscopic systems are saturated with these features. That is why unresolved multiplicity is not ordinarily maintained as lived classical structure.

Principle 21.1 (Classical persistence)**.** *Classical definiteness is the regime in which closure and registration are sufficiently deep that unresolved multiplicity is no longer jointly supportable as persistent realised structure.*

22 Why This Is Not Idealism

Because observer-language is so dangerous here, the paper must state clearly what it is not claiming.

22.1 Not consciousness collapse

The paper does not claim that consciousness creates reality.

22.2 Not observer-exclusive closure

The paper does not claim that closure exists only where there are observers.

22.3 Not human exceptionalism

A human mind is one possible closure-bearing registration frame among others.

22.4 Positive claim

The actual claim is narrower and stronger:

> When a partially structured state couples to a closure-bearing registration frame, the resulting demand for durable record, preserved identity, and downstream consistency acts as an additional bridge across the threshold into persistent realised structure.

That is structural, not mystical.

23 Consequences

Even at this stage, the proposal has several strong consequences.

23.1 It restores a missing middle layer

It restores the missing ontological middle layer between mathematics and actuality: partial structural admissibility.

23.2 It preserves the reality of quantum structure without flattening it into classicality

Superposition is not denied or trivialised. It is given a disciplined place as partial structure near the closure boundary.

23.3 It gives measurement a real but non-magical role

Measurement matters, but not because it mystically produces existence. It matters because record-capable frames alter the supportability conditions of multiplicity.

23.4 It integrates measurement with the broader Darke programme

The measurement problem becomes continuous with the rest of Darke Theory: a problem of closure, admissibility, supportability, and persistence.

23.5 It explains classicality structurally

Classical definiteness is no longer a mysterious separate layer. It is deep closure with thick registrational embedding.

24 Conclusion

The quantum measurement problem is commonly posed as the difficulty of reconciling unitary superposition with definite outcomes. The present paper argues that the deeper issue is better described differently. The real problem is not merely how a wavefunction "collapses", but how formally available multiplicity becomes narrowed into persistent recorded structure.

Darke Theory provides a natural framework for this reformulation because it already distinguishes between formal describability, closure, admissibility, and persistent structure. Within this framework, quantum superposition is interpreted as a regime of partial structure: multiple continuations remain admissible, but not yet fully closure-committed into a single persistent branch. Closure itself is broader than measurement and occurs throughout physical reality. Measurement is therefore not the sole source of closure. It is an additional bridge across the threshold, especially on the side of durable registration.

The observer is correspondingly reinterpreted in structural rather than psychological terms. What matters is not consciousness as such, but the existence of a closure-bearing registration frame capable of preserving outcome, memory, record identity, and downstream consistency. When a partially structured state couples to such a frame, jointly carrying incompatible outcome records becomes structurally unsupportable. Measurement is therefore a closure-selection event: one branch becomes registrationally committed, while alternatives cease to remain jointly supportable as persistent recorded structure.

Under the explicit persistence axioms stated in this paper, measurement in a finite-support closure-bearing registration frame satisfying identity-preserving propagation and event-token uniqueness yields effective outcome singularity at the level of persistent recorded structure.

On this account, collapse is not primitive magic, many-worlds does not settle the ontology simply by preserving mathematical multiplicity, and decoherence remains necessary but not fully sufficient. The decisive transition is the passage from partial structural admissibility to persistent registrational commitment.

The central claim may be restated finally as follows:

> The measurement problem is not best understood as the magical production of actuality by observation, but as the structurally disciplined transition by which

partially admissible multiplicity becomes narrowed into persistent registered structure through closure, preserved identity, and registrational commitment.

That is the claim established here: under the explicit persistence axioms stated in this paper, measurement in a finite-support closure-bearing registration frame yields effective outcome singularity at the level of persistent recorded structure.